

SC6131: Blockchain Innovation Project

Capstone Journal 1

Individual Submission

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⚠ Confidentiality Reminder

This project uses publicly available on-chain data (Etherscan, Dune Analytics) and open-source protocol documentation. No proprietary code, private repository credentials, or NDA-covered material is included.

1. Project Background & Problem Statement

1.1 About the Capstone Partner

The Ethereum Foundation is a non-profit organisation supporting the Ethereum ecosystem through funding, research, and community development. Its research arm actively explores mechanism design for decentralised governance, public goods funding, and capital formation. This Capstone project aligns with the Foundation's "Intelligent Capital Formation" research theme, which examines how on-chain mechanisms can enable more efficient, transparent, and incentive-aligned ways to fund projects, organisations, and collective action.

1.2 Problem Statement

Since 2017, on-chain capital formation on Ethereum has undergone several rounds of mechanism innovation — from undifferentiated ICOs, to reverse Dutch auctions introducing price discovery, to bonding curves enabling continuous pricing, to DAICOs merging DAO governance with fundraising, and finally to Liquidity Bootstrapping Pools (LBPs) achieving fair distribution through dynamic weight adjustment. Despite these innovations, no single mechanism has emerged as a widely accepted standard, and project failures remain common. Our core hypothesis is that these failures stem primarily from an accountability gap: fundraising-stage mechanism design is often sophisticated, but the continuous accountability chain between investors and teams post-fundraising is either absent (ICO, bonding curve), dependent on flawed governance (DAICO), or masked by market liquidity (LBP). This project seeks to empirically verify this hypothesis and, if supported, design a new primitive that addresses it.

1.3 Objectives & Success Criteria

- Objective 1: Establish a project-level dataset covering five major on-chain capital formation mechanisms (ICO, reverse Dutch auction, bonding curve, DAICO, LBP) spanning 2017–2025, with 60–80 representative projects.
- Objective 2: Build and validate an agent-based simulation comparing DAICO and Reverse Dutch Auction under varying behavioral parameters (FOMO, herding), with at least five agent types and Monte Carlo robustness.
- Objective 3: Informed by empirical findings and simulation results, explore design directions for a new accountability primitive that addresses the identified failure modes. Candidate directions include layered accountability mechanisms (combining streaming fund release, individual exit rights, and quadratic governance checkpoints). If a viable concept emerges, prototype it in Solidity.
- Objective 4: Analyse the role of intermediaries (launchpads, market makers, CEX listings) and jurisdictional factors (Howey Test, MiCA, MAS-DPT) in shaping mechanism outcomes.
- Success Criterion: Simulation demonstrates measurable differences in allocation efficiency, manipulation resistance, and long-term incentive alignment across mechanisms; empirical dataset validates or challenges the accountability gap hypothesis; mentor signs off on methodology and design rationale.

2. Technical & Literature Review

2.1 Existing Approaches

- **Reference A:** Buterin, V. (2018). Explanation of DAICOs. Ethereum Research. — Merges ICO with DAO governance via a "tap" mechanism for continuous fund release controlled by token holders. First to introduce post-fundraising investor control. URL: <https://ethresear.ch/t/explanation-of-daicos/465>
- **Reference B:** Heaton & Green (2022). Equitable Continuous Organizations with Self-Assessed Valuations. arXiv:2203.10644. — Identifies the tension between investment efficiency and allocative efficiency in continuous token issuance. URL: <https://arxiv.org/abs/2203.10644>
- **Reference C:** Mechanism Institute. Fundraising Library. — Curated library of on-chain fundraising mechanisms, used as our taxonomy starting point. URL: <https://www.mechanism.institute/library?category=fundraising>
- **Reference D:** Balancer. LBP Documentation. — Describes time-dependent weight shifting to create downward price pressure during token distribution. URL: <https://docs.balancer.fi/concepts/pools/liquidity-bootstrapping.html>
- **Reference E:** Conti et al. (2019). Blockchain-based entrepreneurial finance. JBVI, 12. — Empirical analysis of 253 ICO campaigns, informing our failure mode taxonomy. URL: <https://www.sciencedirect.com/science/article/pii/S0883902619301867>
- **Reference F:** Ethereum Foundation IPTF. Building Private Bonds on Ethereum. — Precedent for encoding accountability mechanisms (periodic coupon obligations) in smart contracts. URL: <https://iptf.ethereum.org/building-private-bonds-on-ethereum/>

2.2 Comparison Table

Approach	Strengths	Limitations for our use case
DAICO (Buterin, 2018)	First to bind fundraising with governance; tap mechanism provides continuous fund control; refund option protects investors	Governance relies on subjective voting — vulnerable to whale manipulation; very few real deployments; teams reluctant to accept constraints
Reverse Dutch Auction	Fair price discovery — price starts high and decays, letting each participant enter at their own valuation; eliminates “first come, first served” problem	One-time fund release post-auction — no post-fundraising accountability; complex mechanism discourages retail participation
Layered Accountability (preliminary direction)	Combines multiple existing primitives (streaming release, individual exit rights, quadratic checkpoints) for defense-in-depth; no single point of failure	Framework-level concept; specific parameter choices require empirical calibration; each layer has known limitations that are mitigated by other layers

2.3 Chosen Technical Direction

The simulation is implemented in Python using a custom agent-based framework with Monte Carlo sampling (N=50 per scenario). The primary mechanism comparison — per supervisor feedback — is DAICO vs. Reverse Dutch Auction, as both mechanisms explicitly address investor protection, making them more commensurable than the original DAICO vs. bonding curve pairing. Behavioral parameters (FOMO intensity, herding coefficient) are incorporated with sensitivity analysis across parameter

sweeps. If a viable accountability primitive emerges from the empirical analysis, a prototype will be implemented in Solidity 0.8.x using Foundry for testing, with a simple frontend demo. This stack was chosen because (a) Python allows rapid iteration on simulation logic, (b) Solidity is the native smart contract language for the Ethereum ecosystem under study, and (c) the supervisor recommended a lightweight parameter sweep over a full ABM framework, which Python handles without heavy dependencies.

3. Scope & Deliverables

3.1 In Scope / Out of Scope

In Scope	Out of Scope
Comparative study of five on-chain capital formation mechanisms (ICO, RDA, bonding curve, DAICO, LBP)	Mainnet deployment of any contract
Agent-based simulation: DAICO vs. RDA with behavioral parameters	Full production-grade ABM framework
Intermediary role analysis (launchpads, market makers, CEX listings)	Production audit or formal verification
Jurisdictional and regulatory comparison (Howey Test, MiCA, MAS-DPT)	Legal advice or regulatory compliance certification
Empirical dataset of 60–80 historical projects	Real-time market data integration
Failure mode taxonomy and accountability gap hypothesis testing	Mobile or consumer-facing application
New accountability primitive design + Solidity prototype (if concept validated)	

3.2 Final Deliverables

- Code repository — simulation framework (Python) + accountability primitive prototype (Solidity, if concept validated) [private GitHub, shared with mentor]
- Public dataset — on-chain capital formation project-level data (60–80 projects, 2017–2025), including intermediary involvement and jurisdictional classification
- Technical design document — accountability primitive specification (informed by empirical findings)
- Final report — Final Report

4. Project Plan & Milestones

Date	Milestone	Deliverable
2026/03/02–03/31	Project kickoff + literature review + scope alignment with mentor	Reference summary + initial mechanism taxonomy
2026/04/01–04/30	Proposal v2 + simulation framework v1–v3	Proposal v2 (English) + simulation code with 5 agent types
2026/05/01–05/17	Simulation refinement + Journal 1 write-up + mentor meeting	Journal 1 (this document)
2026/05/18–07/31	Phase 1 data collection (Dune	Dataset v1.0 + mid-term

	Analytics) + intermediary analysis + empirical analysis + accountability primitive design	analysis report
2026/08/01–10/04	Solidity prototype + stress testing + counterfactual analysis + jurisdictional comparison + Journal 2	Journal 2 + contract prototype
2026/10/05–10/18	Final integration + blog posts (ethresear.ch, Mirror) + Final Report	Final Report + open-source release

5. Work Completed To Date

5.1 Engineering Work

1. Simulation framework (Python, 3 iterations):

- v1 (April): Basic two-mechanism comparison (DAICO vs. bonding curve) with simple rational agents.
- v2 (late April): Updated comparison to DAICO vs. Reverse Dutch Auction per supervisor feedback; added FOMO and herding behavioral parameters.
- v3 (May): Full framework with 5 agent types (rational long-term 40%, speculator 20%, whale 10%, FOMO follower 20%, late entrant 10%), per-agent token tracking, Gini coefficient for allocation fairness, top-10% concentration metric, price volatility metric, and Monte Carlo N=50 per scenario across 7 scenarios (FOMO sweep + herding combinations). Code: capstone_sim_v3.py.

2. Key simulation results:

Scenario	DAICO Efficiency	RDA Funds Raised
Baseline (no FOMO)	51%	133.72
FOMO = 0.3	52%	~144
FOMO = 0.7	54%	~144
FOMO + Herding	53%	~144

Key findings:

- FOMO improves RDA allocation fairness (Gini 0.515 → 0.401): counterintuitively, FOMO drives more participants to enter early at higher prices, spreading tokens more evenly rather than concentrating them in a few early buyers.
 - DAICO is FOMO-invariant: Gini stays at ~0.491 regardless of behavioral parameters, because DAICO contribution happens at the start and is not affected by price discovery dynamics.
 - DAICO governance is robust in simulation: 0% destroy rate across all scenarios (current threshold: 15% of agents voting destroy).
 - Herding effect is secondary: adding herding changes results marginally compared to FOMO alone.
3. **Proposal v2 → v3:** Iteratively updated per supervisor feedback — v2 changed primary comparison to DAICO vs. RDA and added FOMO/herding parameters; v3 further refined the metrics framework definition (noted as independent research contribution), elaborated the layered accountability architecture direction, and detailed next-phase plans. File: capstone_proposal_v3.md.

5.2 Research & Design Work

- Reviewed 7+ references covering all five mechanisms (see Section 2 and Reference list).
- Drafted failure mode taxonomy: exit scam, soft rug, governance failure, liquidity trap, slow death, regulatory shutdown.
- Began exploring accountability primitive design directions. Initial explorations identified fundamental challenges faced by any single accountability mechanism: oracle dependency for subjective milestones, rigidity vs. development iteration needs, and adoption barriers similar to DAICO. Currently shifted to evaluating a layered accountability approach combining streaming fund release, individual exit rights (Moloch-style rage quit), and quadratic governance checkpoints, using multi-layer composition to raise bypass costs. Final design direction to be determined after Phase 1 empirical validation.
- Prepared meeting presentation script with Q&A responses (used during 05/11 mentor meeting).
- Mentor confirmed (05/11): endorsed proposed next steps and methodology direction.

5.3 Initial Data Collection (Pilot Batch)

Compiled an initial dataset of 50 projects across all five mechanism types (ICO: 10, RDA: 10, bonding curve: 10, DAICO: 10, LBP: 10) from public sources (CryptoRank, Messari, Dune Analytics, project documentation). Fields collected: project name, mechanism type, year, amount raised, token, current status, and notes. Key early observation: DAICO has the fewest real implementations (only 1–2 verified deployments) despite strong theoretical design, supporting the accountability gap hypothesis. Full dataset expansion and on-chain validation planned for Phase 1.

5.4 What Is NOT Done Yet

- Full on-chain data validation and expansion (target: 60–80 projects with Dune Analytics on-chain metrics).
- Full 3×3×3 scenario matrix (team integrity × market conditions × attacker presence — 27 scenarios).
- Secondary market simulation (post-auction trading dynamics).
- Accountability primitive design finalisation (informed by empirical data).
- Solidity prototype (if concept validated).
- Counterfactual analysis on historical failed projects.
- Intermediary role analysis and jurisdictional comparison.
- ethresear.ch and Mirror blog posts.

6. Mentor Communication Log

Date	Format	Topics Discussed	Action Items / Owner
2026/03/05	Meeting	Project kickoff introduction + reading materials distributed	Student: review distributed reference materials, familiarise with research direction
2026/03/15	Email	Initial project brief + reference materials shared by mentor	Student: review references (DAICO paper, Continuous Org paper, Mechanism Institute library, ScienceDirect paper,

			Ethereum bonds paper)
2026/04/17	Email	Student asked 3 scoping questions: (1) head-to-head vs. broad survey, (2) behavioral realism in simulation, (3) evaluation metrics benchmarks	Mentor: responded with feedback on all three
2026/04/30	Email	Mentor feedback on proposal v1: change comparison to DAICO vs. RDA; add FOMO/herding; define own metrics framework	Student: update proposal v2 + simulation (done)
2026/05/03	Email	Student shared progress update + requested meeting	Mentor: scheduled meeting for 05/11
2026/05/11	Zoom	Presented simulation v3 results (FOMO finding, DAICO stability); discussed data collection plan; confirmed next steps	Mentor confirmed direction; Student: begin Dune Analytics data collection; expand scenario matrix

7. Risks & Blockers

- Risk: Dune Analytics query complexity — some historical projects (2017–2018 era) may have limited on-chain data availability. Mitigation: Supplement with CryptoRank, Messari, and manual Etherscan investigation; reduce minimum sample per mechanism to 8 if needed.
- Risk: Simulation model ceiling — current 30-agent model saturates quickly (e.g., all 30 participate under high FOMO). Mitigation: Scale to 100+ agents for Phase 2; add secondary market dynamics.
- Risk: Accountability primitive design complexity — layered approach requires careful parameter tuning and interaction analysis between layers. Mitigation: Use simulation framework to test layer interactions before prototyping; start with simplest viable configuration.
- Blocker: None currently. Mentor has confirmed direction and next steps are clear.
- Help needed from NTU: Access to NTU library databases for full-text access to academic references (some papers behind paywalls); potential introduction to industry contacts for intermediary analysis (launchpad operators, market makers).

8. Plan for the Next Phase (2026/05/18 – 2026/10/04)

- May–July (Phase 1, milestone: 2026/05/18–07/31): On-chain data collection via Dune Analytics (20–30 projects), validate simulation against real data, map intermediary roles, expand to full 27-scenario matrix, finalise accountability primitive direction.
- August–early October (Phase 2, milestone: 2026/08/01–10/04): Solidity prototype with Foundry, counterfactual analysis on 5–10 failed projects, stress testing, jurisdictional comparison (Howey/MiCA/MAS-DPT). Journal 2 due 10/04.

- Mid-October (Phase 3, milestone: 2026/10/05–10/18): Final integration. Prepare ethresear.ch and Mirror blog posts. Write Final Report (due 10/18). Open-source release.

9. Personal Reflection

The most valuable lesson from this first phase has been learning how to translate a broad research question into a testable simulation. When I started, the project scope felt overwhelming — five mechanisms, intermediary analysis, jurisdictional considerations, and a new primitive design. The supervisor’s feedback to narrow the primary comparison to DAICO vs. Reverse Dutch Auction was a turning point: it forced me to think about what makes two mechanisms commensurable rather than trying to compare everything at once.

The most surprising finding so far has been the FOMO result. My intuition was that behavioral effects like FOMO would distort outcomes and make mechanisms perform worse. Instead, the simulation showed that FOMO actually improves allocation fairness in reverse Dutch auctions by drawing in more participants. This reminded me that mechanism design operates in a world of real human behavior, not rational agents alone — a lesson I plan to carry into the accountability primitive design phase.

If I were starting over, I would have begun the simulation work earlier rather than spending the first month primarily on literature review. Having concrete results made the mentor conversations much more productive — the 05/11 meeting was the first time I could show actual data, and the mentor’s engagement was noticeably higher. For the next phase, I plan to maintain this “build first, refine later” approach.

Appendix

- A. Repository link: Private GitHub repository (shared with mentor and supervisor). Contains: `capstone_sim_v3.py` (simulation framework), `capstone_proposal_v3.md` (proposal), analysis notebooks.
- B. Annotated bibliography: See Section 2.1 for detailed reference summaries.
- C. Simulation output sample: See Section 5.1 for summary table. Full Monte Carlo output available in repository.
- D. Architecture direction: Layered Accountability Framework (preliminary): (1) Streaming Fund Release (Sablier-style) → (2) Individual Exit Rights (Moloch-style rage quit) → (3) Quadratic Governance Checkpoints → (4) On-chain Signal Monitor. Detailed architecture and interaction design to be developed in Phase 2 after empirical validation.

— End of Journal 1 —